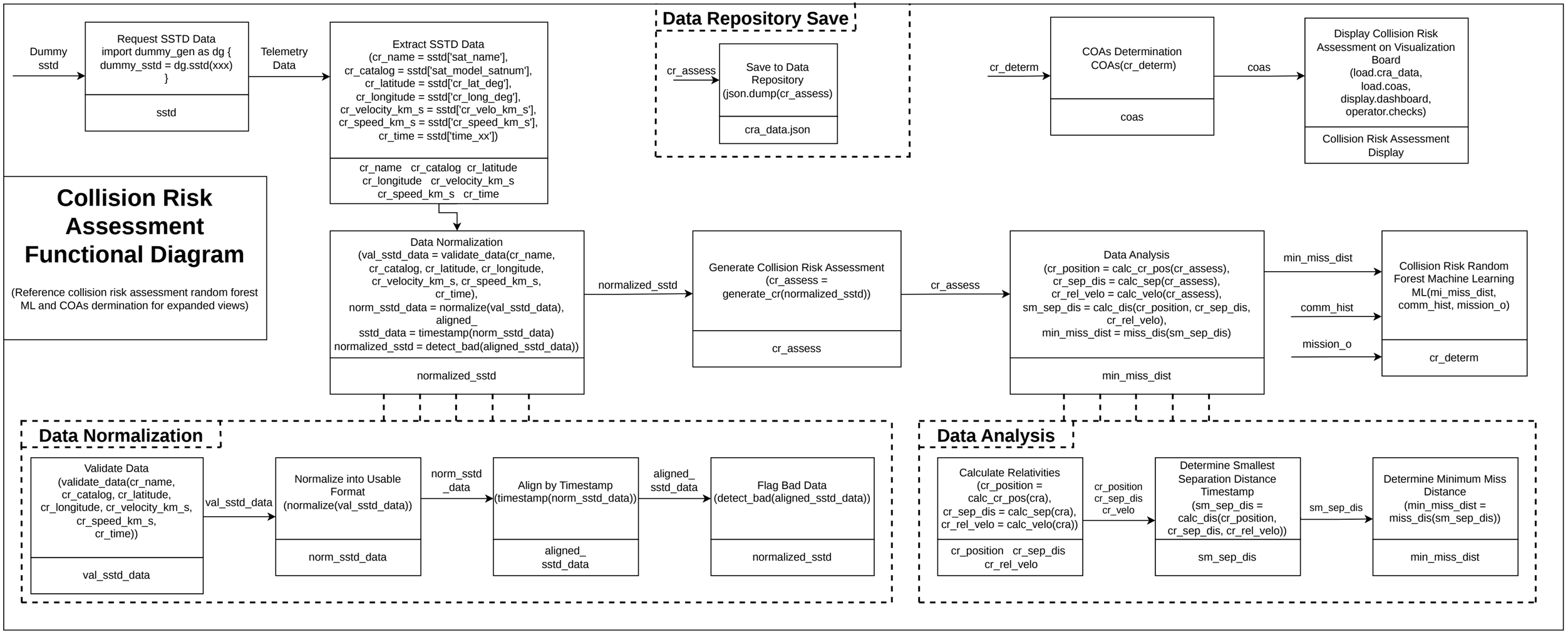
Collision Risk Assessment Functional Diagram
(Reference collision risk assessment random forest ML and COAs dermination for expanded views)
Dummy sstd
Request SSTD Data
import dummy_gen as dg {
dummy_sstd = dg.sstd(xxx)
}
sstd
Telemetry Data
Extract SSTD Data
(cr_name = sstd['sat_name'],
cr_catalog = sstd['sat_model_satnum'],
cr_latitude = sstd['cr_lat_deg'],
cr_longitude = sstd['cr_long_deg'],
cr_velocity_km_s = sstd['cr_velo_km_s'],
cr_speed_km_s = sstd['cr_speed_km_s'],
cr_time = sstd['time_xx'])
cr_name cr_catalog cr_latitude
cr_longitude cr_velocity_km_s
cr_speed_km_s cr_time
Data Repository Save
cr_assess
Save to Data Repository
(json.dump(cr_assess)
cra_data.json
cr_determ
COAs Determination
COAs(cr_determ)
coas
coas
Display Collision Risk Assessment on Visualization Board
(load.cra_data,
load.coas,
display.dashboard,
operator.checks)
Collision Risk Assessment Display
Data Normalization
(val_sstd_data = validate_data(cr_name,
cr_catalog, cr_latitude, cr_longitude,
cr_velocity_km_s, cr_speed_km_s,
cr_time),
norm_sstd_data = normalize(val_sstd_data),
aligned_
sstd_data = timestamp(norm_sstd_data)
normalized_sstd = detect_bad(aligned_sstd_data))
normalized_sstd
normalized_sstd
Generate Collision Risk Assessment
(cr_assess =
generate_cr(normalized_sstd))
cr_assess
cr_assess
Data Analysis
(cr_position = calc_cr_pos(cr_assess),
cr_sep_dis = calc_sep(cr_assess),
cr_rel_velo = calc_velo(cr_assess),
sm_sep_dis = calc_dis(cr_position, cr_sep_dis,
cr_rel_velo),
min_miss_dist = miss_dis(sm_sep_dis)
min_miss_dist
min_miss_dist
comm_hist
mission_o
Collision Risk Random Forest Machine Learning ML(mi_miss_dist,
comm_hist, mission_o)
cr_determ
Data Normalization
Validate Data
(validate_data(cr_name,
cr_catalog, cr_latitude,
cr_longitude, cr_velocity_km_s,
cr_speed_km_s,
cr_time))
val_sstd_data
val_sstd_data
Normalize into Usable Format
(normalize(val_sstd_data))
norm_sstd_data
norm_sstd_data
Align by Timestamp
(timestamp(norm_sstd_data))
aligned_
sstd_data
aligned_sstd_data
Flag Bad Data
(detect_bad(aligned_sstd_data))
normalized_sstd
Data Analysis
Calculate Relativities
(cr_position =
calc_cr_pos(cra),
cr_sep_dis = calc_sep(cra),
cr_rel_velo = calc_velo(cra))
cr_position cr_sep_dis
cr_rel_velo
cr_position
cr_sep_dis
cr_velo
Determine Smallest Separation Distance Timestamp
(sm_sep_dis =
calc_dis(cr_position,
cr_sep_dis, cr_rel_velo))
sm_sep_dis
sm_sep_dis
Determine Minimum Miss Distance
(min_miss_dist =
miss_dis(sm_sep_dis))
min_miss_dist